

Two-Link Inverse Kinematics (Project 1)

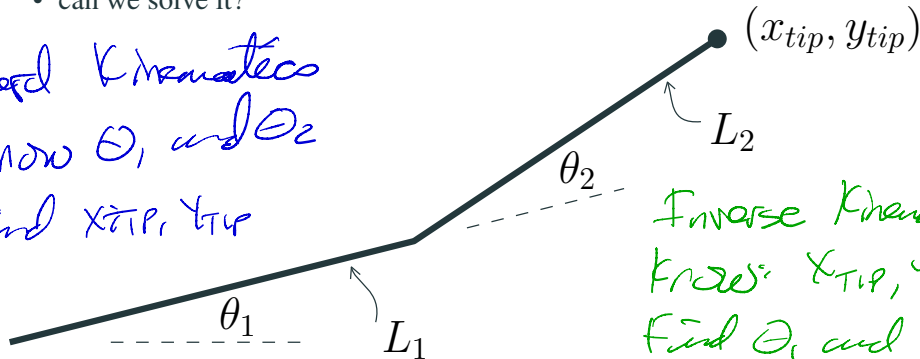
EGR 445/545

GVSU

Two-Link Inverse Kinematics: Problem Statement

- what is known? $\rightarrow x_{tip}, y_{tip}, L_1$ and L_2
- what is unknown? $\rightarrow \theta_1$ and θ_2
- can we write a relationship between $x_{tip}, y_{tip}, \theta_1$, and θ_2 ?
- can we solve it?

Forward Kinematics
Know θ_1 and θ_2
Find x_{tip}, y_{tip}



Inverse Kinematics
Know: x_{tip}, y_{tip}
Find θ_1 and θ_2

Known symbolic relationship

$$x_{Tip} = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

$$y_{Tip} = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

Two-Link Inverse Kinematics: α , β , r , and ψ ?

- can we find α , β , r and ψ ?
- would finding them help?

$$r^2 = x_{tip}^2 + y_{tip}^2$$

$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos \alpha$$

α is the angle opposite of r

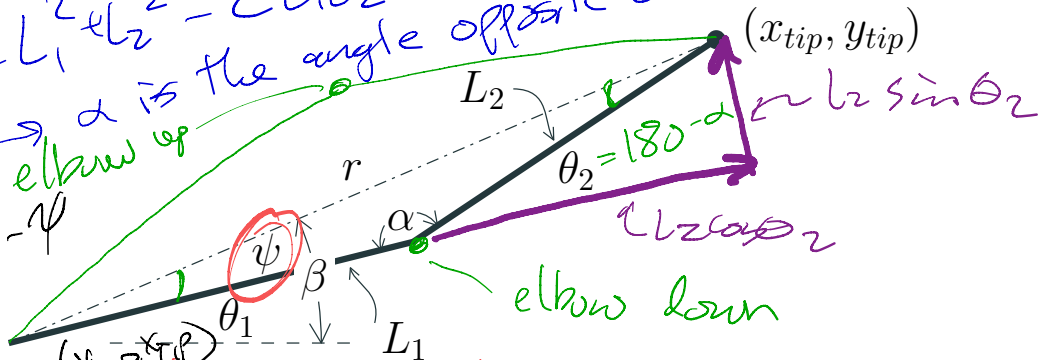
elbow up

$$\theta_1 = \beta - \psi$$

$$\beta = \arctan2(y_{tip}, x_{tip})$$

$$\psi = \arctan2(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$$

elbow down



$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos \alpha$$

$$2L_1L_2 \cos \alpha = L_1^2 + L_2^2 - r^2$$

$$\cos \alpha = \frac{L_1^2 + L_2^2 - r^2}{2L_1L_2}$$

$$\alpha = \arctan 2(\sin \alpha, \cos \alpha)$$

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\sin \alpha = \pm \sqrt{1 - \cos^2 \alpha}$$

Two-Link Inverse Kinematics: ~~Document Camera~~

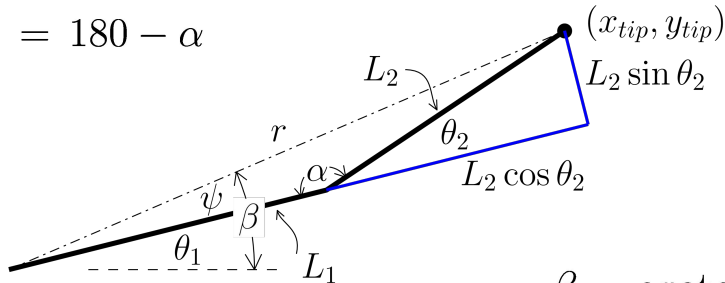
- let's do this...

Two-Link Inverse Kinematics Summary

$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(\alpha)$$

$$\theta_2 = 180 - \alpha$$

$$r^2 = x_{tip}^2 + y_{tip}^2$$



$$\beta = \arctan2(y_{tip}, x_{tip})$$

$$\psi = \arctan2(L_2 \sin(\theta_2), L_1 + L_2 \cos(\theta_2)) \quad \theta_1 = \beta - \psi$$